

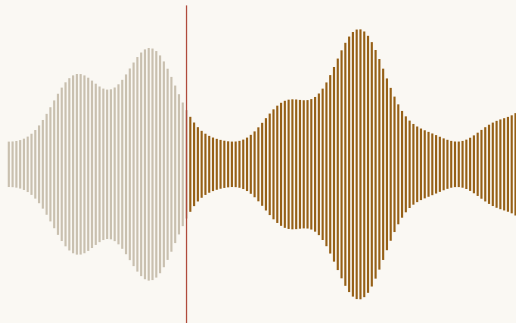
SonicCraft

Hear the change.
See the signal.

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Python desktop audio editor



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Motivation and problem

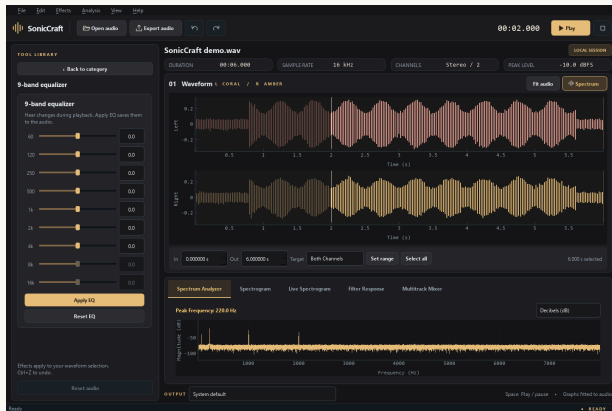
Connect signal-processing concepts to changes we can hear.

MOTIVATION

Learn DSP by listening and experimenting.

PROBLEM ADDRESSED

Noise and uneven levels make recordings harder to use.



Overall approach

A repeatable workflow: inspect, preview, commit and compare.

Inspect

Open audio, select a region and examine its signal.

Preview

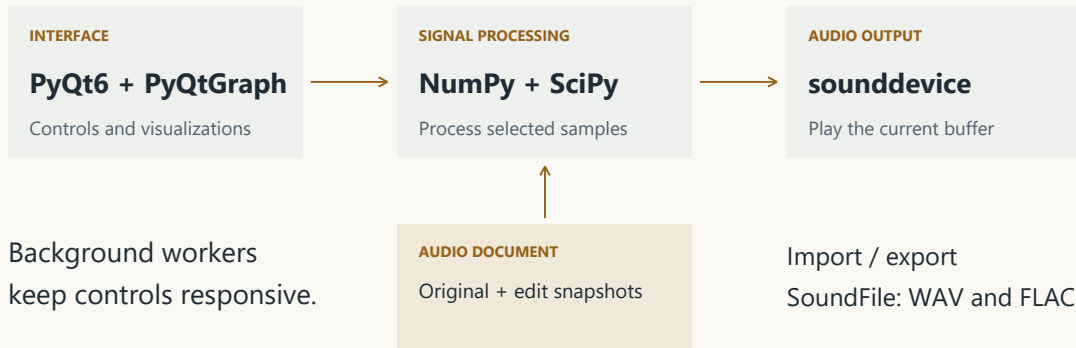
Adjust a tool and hear the effect during playback.

Commit and compare

Apply the edit, compare with the original, then export.

Architecture and design

One audio document connects editing, processing and playback.



Relevant theory and concepts

The same signal can be studied in time and frequency.

Samples and amplitude

Gain scales samples; fades change their level over time.

Fourier analysis and filters

FFT reveals frequency content; filters and EQ reshape it.

Windowed analysis

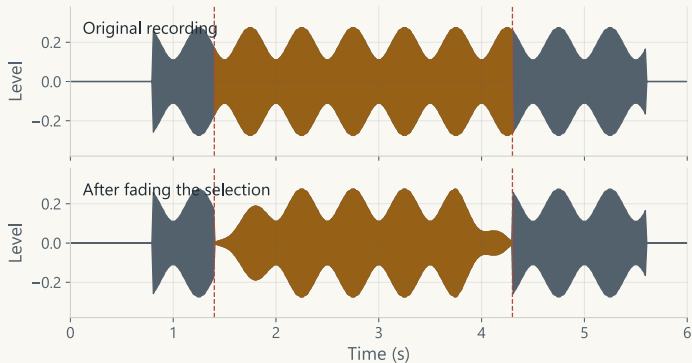
Short FFT windows form a spectrogram and guide noise reduction.

Edit & Dynamics

Shape the selected part of a recording.

EDIT & DYNAMICS

- Select, trim or remove a time range.
- Adjust level or normalize the peak.
- Fade edges for a smooth entrance or exit.



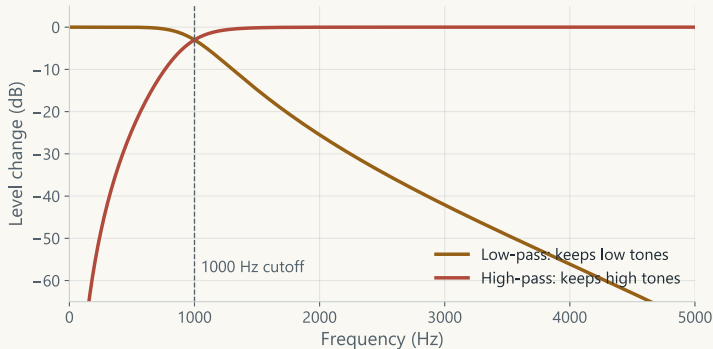
A real clip: the highlighted interval fades at both ends.

Frequency filter

Keep wanted frequencies and reduce others.

FILTERS & EQUALIZER

- Choose low-pass, high-pass or band filters.
- Set the cutoff to shape what passes.
- Hear the result while adjusting.



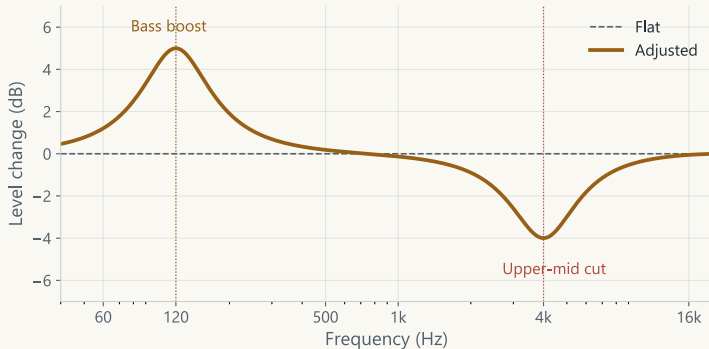
Two filter settings at the same cutoff frequency.

9-band equalizer

Change the tonal balance of the recording.

FILTERS & EQUALIZER

- Move a band to boost or reduce its range.
- Combine bands across bass, mids and treble.
- Apply EQ to save what you hear.



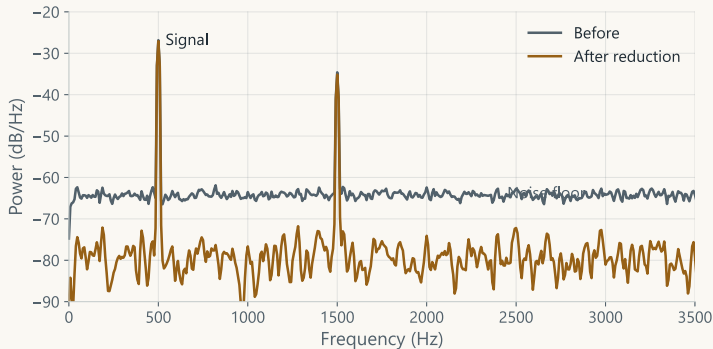
Example: bass boost and upper-mid cut; other bands stay flat.

Noise reduction

Reduce steady background sound.

NOISE REDUCTION

- Capture a section containing only noise.
- Estimate its frequency pattern.
- Reduce matching noise in the selection.



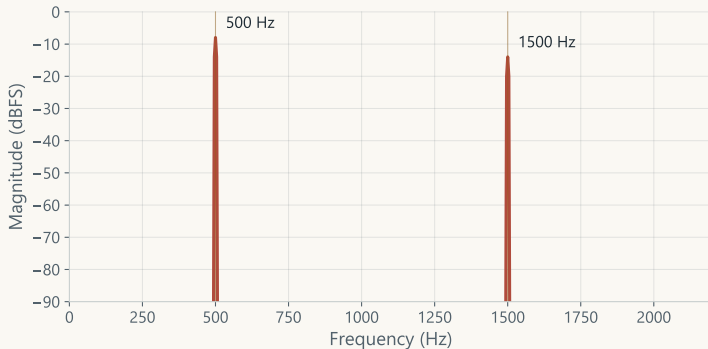
The noise floor drops while two signal peaks remain.

Spectrum analyzer

See which frequencies exist in the selection.

SPECTRUM ANALYZER

- Select audio and calculate its FFT.
- Read the strongest frequency peaks.
- Compare the spectrum before and after edits.



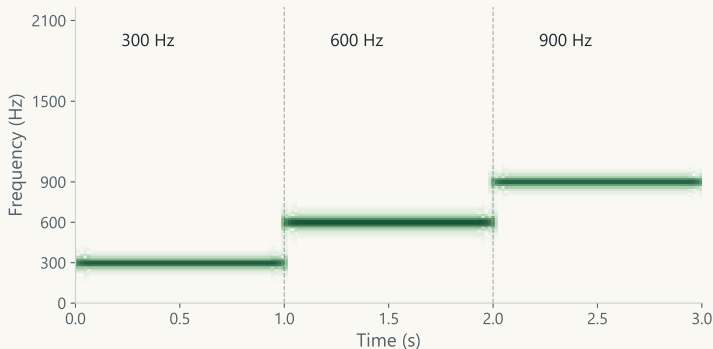
A two-tone example produces peaks at 500 Hz and 1500 Hz.

Spectrogram

See how frequency content changes with time.

SPECTROGRAM

- Time runs left to right; frequency runs upward.
- Green bands show strong components.
- Find when a tone or sound occurs.



Three notes played in sequence: 300, 600 and 900 Hz.

Live spectrogram

Real Time Spectrogram Generation

LIVE SPECTROGRAM

- Analyze audio output in real time.
- Observe spectral changes across time and frequency.
- Adjust FFT window parameters dynamically without stopping playback

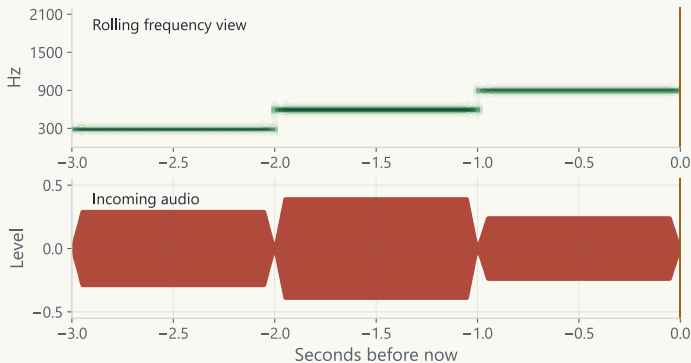
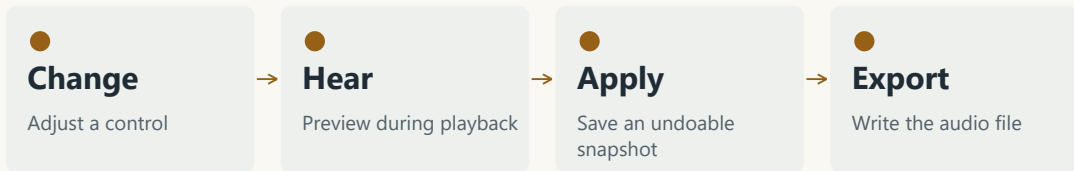


Illustration with generated notes; the live view uses microphone input.

From preview to finished audio

Final result: inspect, edit, hear and export in one desktop application.



TWO RESET SCOPES

Tool reset

Clears this tool's settings and preview.

Reset audio

Restores the original audio and all settings.

SONICCRAFT

Explore the project in code.

Source, documentation and a reproducible demo

github.com/mehrab4226/SonicCraft